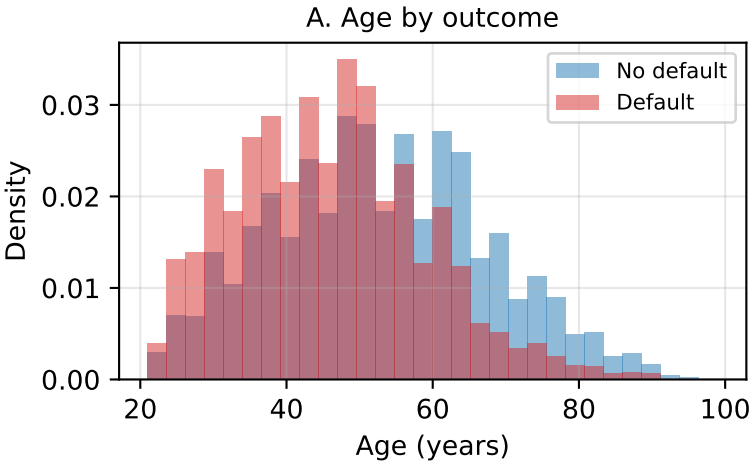
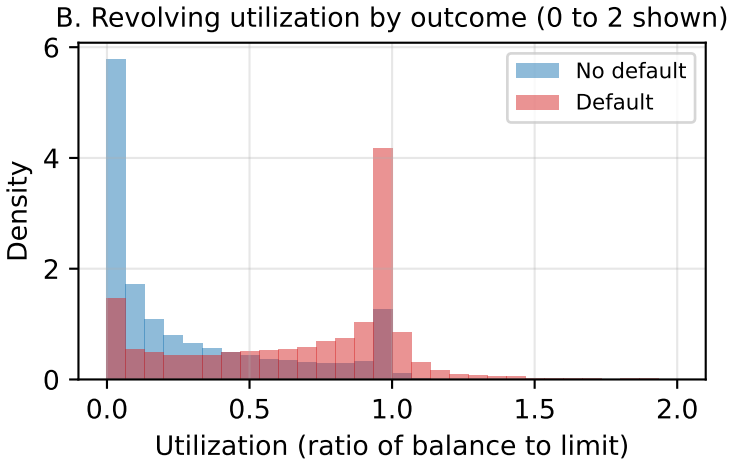
# Credit Default Risk: Feature Distribution Analysis

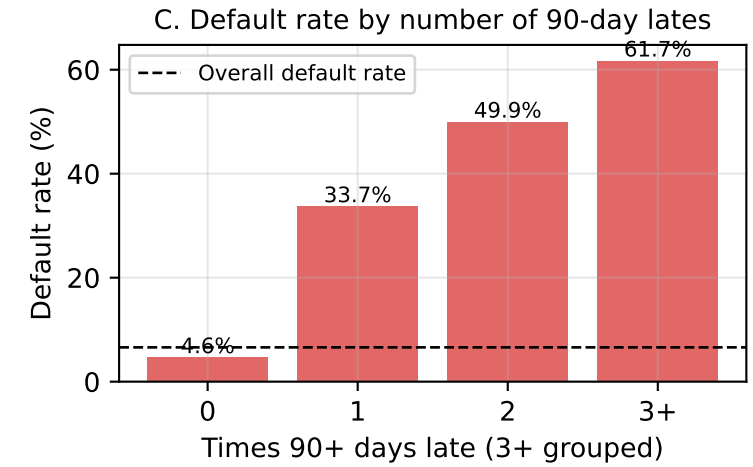
Give Me Some Credit dataset | 149,710 customers after cleaning | overall default rate 6.6% | blue = no default, red = default

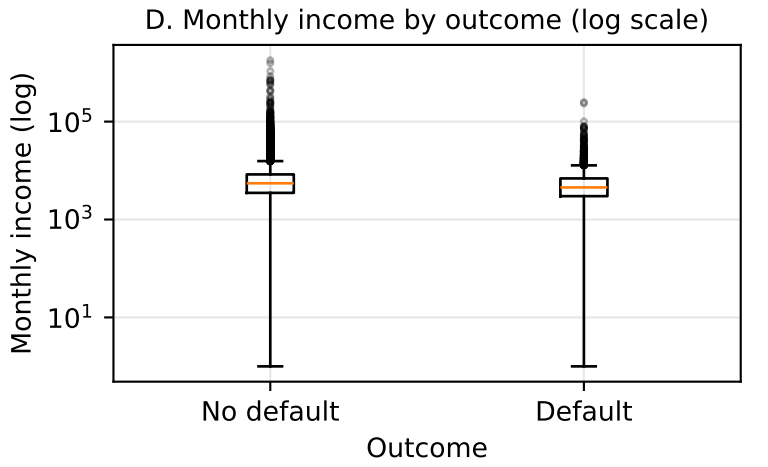



Age is roughly symmetric (skewness 0.19) with 0% missing values, 0% zeros and only 0.02% IQR outliers. Defaulters are younger: mean age 46.1 vs 52.8 years, and the default rate falls steadily from 11.2% (under 30) to 2.3% (70+). Credit-risk reading: younger borrowers usually have a shorter credit history, less stable income and smaller savings buffers, while older borrowers tend to be more established.

Utilization is extremely right-skewed (skewness 68): median 0.15, maximum 29,110; 2.2% of customers are above 1.0 and 0.16% above 10 (likely data errors). Outliers make the means misleading (5.8 vs 4.4); the medians show the real gap: 0.13 (no default) vs 0.83 (default). The default rate rises steadily with utilization, from 2.1% below 0.25 to 37.3% above 1.0 (17x), and 31% of defaulters sit right at their limit (0.95-1.05) vs 9% of non-defaulters. Hypothesis: customers using most of their limit are financially stretched and have little room for a shock.




The feature is dominated by zeros (94.6%) and is strongly right-skewed (maximum 17). At least one 90-day delinquency is rare (5.4% of customers), but the default rate is 41.2% for those customers vs 4.6% for the rest (8.9x higher). It is a strong signal when present, but not enough alone: 66.3% of defaulters have no 90-day late on record (vs 96.6% of non-defaulters), so it identifies a minority of defaulters very well and misses most of them. Past severe delinquency reflects repayment behaviour.

Income is right-skewed (skewness 75.5) with 19.8% missing values and 1.1% zeros; 4.1% of known values are IQR outliers (maximum 1,794,060). The median is lower for defaulters (4,500 vs 5,470), but the gap is modest. Customers with missing income default less often (5.4% vs 6.9%), so the missingness itself carries information and should be kept as a flag. Hypothesis: lower income means a thinner buffer for repayments.

## Final insights: three starting features

I would start with RevolvingUtilizationOfUnsecuredLines (IV 1.09), NumberOfTimes90DaysLate (IV 0.85) and age (IV 0.25). Utilization shows current credit stress (default rate 2.1% to 37.3% across levels) and 90-day lates show past repayment behaviour (41.2% vs 4.6%), while age adds a different, easy-to-collect signal (11.2% for under 30 vs 2.3% for 70+). MonthlyIncome and DebtRatio are weaker (IV 0.07 and 0.08) and have missing or extreme values.

Notes: EDA on a cleaned copy of the data (leaked column NumberOfDefault removed, ages outside 18-100 removed); the original CSV was not modified. IV = Information Value. Outliers = values outside 1.5 x IQR.